

Towards Realistic Conversational Head Generation: A Comprehensive Framework for Lifelike Video Synthesis

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ABSTRACT

The Vivid Talking Head Video Generation track of the “ACM Multimedia ViCo 2023 Conversational Head Generation Challenge” aims to generate realistic face-to-face conversation videos based on audio and reference images. However, the direct synthesis of reference videos from audio and reference images poses a significant challenge. In response, we propose a comprehensive method that combines audio-driven 3DMM parameter prediction with a rendering network to generate high-fidelity lip-sync talking head videos. In the first stage, we leverage the audio input to predict the 3DMM parameters, capturing essential facial expressions and head poses needed for realistic video generation. In the second stage, we employ a sophisticated rendering network designed to ensure accurate lip synchronization and natural facial expressions. This network enhances the visual quality and realism of the generated videos. We are delighted to announce that our method achieved first place in the talking head generation track of the challenge and was honored with the People’s Selection Award. The source code for our method is accessible¹, allowing others to build upon our work and drive further advancements in the field of conversational head generation.

CCS CONCEPTS

• Information systems → Multimedia content creation.

KEYWORDS

Vivid Talking Head Generation, Conversational Head Generation

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¹https://github.com/sdulyq/ACM2023_workshop.

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1 INTRODUCTION

One-shot talking head generation, a rapidly advancing research field, aims to generate videos of individuals speaking, exhibiting natural expressions and movements in sync with given audio input, from a single facial image [25]. This technology has been adopted in diverse applications ranging from online education and video conferencing [5] to digital humans and the Metaverse.

The ViCo conversational head generation challenge [30], hosted in conjunction with ACM Multimedia 2023, includes two tasks: 1) generating an animated talking head, driven by a reference image and audio, and 2) synthesizing a listening head based on the reference image and speaker’s behaviors. Our work is primarily focused on the track of vibrant talking head video generation, conditioned on the 3DMM parameters. While it is possible to create realistic digital speakers by generating numerous videos from a single speaker [20], there remains a critical need to explore the generation of lifelike digital speakers from limited data.

In this paper, we aim to extract accurate and natural 3DMM parameters from limited data and develop a robust rendering network to generate vibrant videos. With these goals in mind, we propose a two-stage framework to tackle the task of one-shot talking head generation. In the first stage, we introduce two separate neural network modules: the natural head pose parameter generator and the precise expression parameter generator. These are designed to extract natural head pose parameters and precise expression parameters from limited data, respectively. By harnessing the detailed information that these parameters offer, we can better capture the intricate dynamics of speaking behaviors. In the second stage, we utilize a robust rendering network to translate the predicted 3DMM parameters into a series of vibrant images. The experimental results not only validate the superiority of our method in terms of overall performance but also underline its proficiency in generating high-quality images, natural head poses, and precisely synchronized facial expressions.

2 RELATED WORK

Audio-driven talking head generation can be classified into two main categories based on the type of visual input: few-shot talking head generation and one-shot talking head generation.

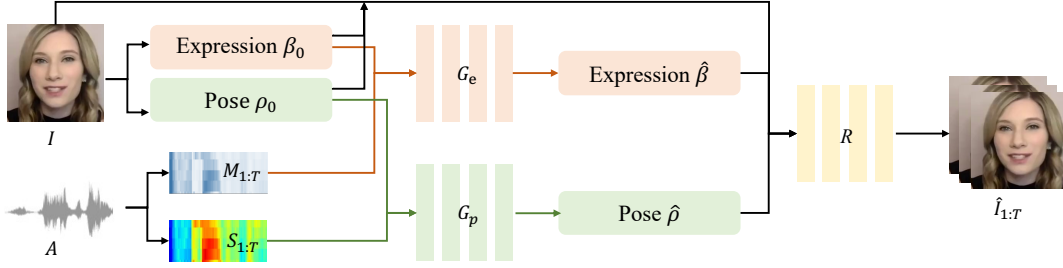


Figure 1: The framework of our proposed model.

2.1 Few-shot Talking Head Generation

The process of few-shot talking head generation involves using a video of the target identity’s talking head to capture emotions, movements, and other visual information. Early research primarily concentrated on video dubbing, which entailed altering the mouth movements of input videos based on the given audio. With the successful integration of Generative Adversarial Networks (GANs) in image generation, later works such as [13, 14] directly extract and synthesize mouth movements using GANs. However, alterations restricted to the mouth region often led to unnatural facial movements and noticeable discontinuities at the splice.

Inspired by deep video portraits [9], certain approaches employ the 3D Morphable Model (3DMM) as an intermediary representation to adjust facial expressions in talking head videos based on audio. Efforts like AudioDVP [23] and AD-NeRF [6] reconstruct expression parameters to manipulate mouth shape but still depend on lengthy input videos to coincide with the audio duration, resulting in protracted generation times. In contrast, other methods extract particular features from short videos to generate new videos [17][12]. For instance, Song *et al.* [17] obtained precise 3DMM expression parameters from reference videos and updated them based on the input audio. Pang *et al.* [12] concentrated on replicating head motions from input videos to create natural head poses. However, in real-world applications, obtaining high-quality talking head videos of the target individual can prove challenging. Consequently, an increased focus has been directed toward the one-shot talking head generation.

2.2 One-shot Talking Head Generation

One-shot talking head generation focuses on creating motion from a single image. Unlike its few-shot counterpart, this task is centered around animating one standalone image. Earlier works [8, 18, 28, 33] heavily lean on powerful lip-sync discriminators to produce high-quality talking head generations using GANs. However, the training of GANs is notoriously difficult, often leading to mode collapse, rendering them less suitable for this task. As a result, some research [1, 16, 19, 24] pivot to the utilization of diffusion models for the generation of high-quality talking head videos. Despite the impressive results of videos generated by diffusion models, they are hindered by low generation efficiency. In this paper, we sidestep diffusion models and instead incorporate the 3DMM as an intermediate representation for synthesizing talking head videos from the audio modality.

Prior endeavors in one-shot methodologies have integrated intermediate representations to achieve their objectives [21, 29, 32].

A subset of these works has probed into the adept representation capabilities intrinsic to the 3DMM [10, 15, 26, 27]. For example, [26] and [10] have employed modifications in the expression and pose parameters of the benchmark image, subsequently rendering the refined 3DMM to video frames. However, these methodologies, while pioneering, still grapple with challenges encompassing imprecise lip articulations, discernible visual aberrations, and less-than-optimal rendering fidelity.

3 METHODOLOGY

3.1 Problem Formulation

Given the auditory input A and a reference image I of the speaker, our aim is to generate a synchronized high-resolution video V . As outlined in Figure 1, our proposed model bifurcates into two main phases: prediction of the 3DMM parameters and rendering. During the 3DMM parameter prediction phase, we make use of the Natural Head Pose Parameter Generator G_p and the Precise Expression Parameter Generator G_e to prognosticate parameters corresponding to the head pose and facial expressions, respectively. These parameters are subsequently processed through an established rendering network R , leading to the generation of image frames that constitute the final video.

3.2 Natural Head Pose Parameter Generator G_p

Initially, the raw auditory input, denoted as A , undergoes segmentation into an auditory sequence $a_{1:T}$, aligned with the frames per second (fps) intrinsic to the associated video. Within this context, a_i stands representative of an individual audio frame. Subsequent to this partitioning, the auditory features, designated as $S_{1:T}$, are extracted employing the methodology prescribed by ViCo². Each distinct feature, S_i encapsulates Mel Frequency Cepstral Coefficients (MFCC), supplemented by corresponding MFCC Delta and DeltaDelta attributes, energy, amplitude, and the zero-crossing rate.

To representatively capture the speaker’s motions, we enlist the 3DMM parameters. Specifically, the 64-dimensional parameter β is tailored to expressions, whereas the 6-dimensional parameter ρ attends to head motions. Leveraging a widely-accepted tool [4], we derive the foundational parameters β_0 and ρ_0 from the reference image I . The formulaic articulation of deriving the projected head parameters $\hat{\rho}$ is illustrated as $\hat{\rho}_{1:T} = G_p(\rho_0, S_{1:T})$.

Despite the adoption of an architectural approach akin to the precedent work [10], we have also embedded alterations. Given the one-to-many mapping from audio to head pose parameters,

²https://github.com/dc3ea9f/ViCo_challenge_baseline/

Table 1: Quantitative performance comparison between our model and other baselines.

Rank	Team Name	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	PoseL1↓	ExpL1↓	AVOffset→ 0	AVConf↑	LipLMD↓
1	ilearn	0.599	0.207	17.279	28.665	0.500	0.089	0.149	-1.378	3.291	11.143
2	Robo Space	0.613	0.203	17.822	29.163	0.546	0.101	0.147	-0.867	2.564	12.147
3	metah	0.613	0.204	17.811	28.829	0.540	0.101	0.151	-1.733	2.541	12.192
4	USTC-IAT-United	0.576	0.206	16.757	13.622	0.517	0.090	0.164	-0.311	0.273	11.982

Table 2: Performance comparison among the variants of our model.

Variants	SSIM↑	CPBD↑	PSNR↑	FID↓	CSIM↑	PoseL1↓	ExpL1↓	AVOffset→ 0	AVConf↑	LipLMD↓
w/o G_p	0.602	0.200	17.397	27.606	0.522	0.097	0.150	-0.622	2.554	12.390
w/o R	0.552	0.192	16.195	33.986	0.506	0.091	0.151	-1.511	2.045	11.079
Full Model	0.599	0.207	17.279	28.665	0.500	0.089	0.149	-1.378	3.291	11.143

the weight of the L2 loss between the factual ρ and the projected $\hat{\rho}$ is lessened. This adjustment is conducted in reference to Audio2Head [21], which allows for the incorporation of the Structural Similarity (SSIM) loss. The losses involved in the prediction procedure can be mathematically expressed as:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_p + \lambda_2 \mathcal{L}_{adja} + \lambda_3 \mathcal{L}_{SSIM}, \quad (1)$$

where λ_1 , λ_2 , and λ_3 symbolize pre-determined balance parameters.

Within this framework, $\mathcal{L}_{adja}$ signifies the inter-frame smoothness loss, mathematically represented as:

$$\mathcal{L}_{adja} = \sum_{t=1}^T \|v(\rho_t) - v(\hat{\rho}_t)\|_2, \quad (2)$$

where $v(\cdot)$ symbolizes the inter-frame variations between the contemporary frame and its preceding adjacent frame.

$\mathcal{L}_p$ indicates the L2 loss between the projected and the factual values, mathematically represented as:

$$\mathcal{L}_p = \sum_{t=1}^T \|\rho_t - \hat{\rho}_t\|_2. \quad (3)$$

$\mathcal{L}_{SSIM}$ denotes the SSIM loss between the projected and the factual values, mathematically defined as:

$$\mathcal{L}_{SSIM} = 1 - \frac{(2\mu\hat{\mu} + C_1)(2cov + C_2)}{(\mu^2 + \hat{\mu}^2 + C_1)(\sigma^2 + \hat{\sigma}^2 + C_2)}, \quad (4)$$

where $\hat{\mu}$ and $\hat{\sigma}$ stand for the mean and standard deviation of $\hat{\rho}_{1:T}$, respectively. Conversely, μ and σ correspond to the mean and standard deviation of the factual head pose sequence $\rho_{1:T}$. cov signifies covariance while C_1 and C_2 are minor constants.

3.3 Precise Expression Parameter Generator G_e

Drawing inspiration from the laudable accomplishments of SadTalker in synthesizing highly congruent videos [26], we integrate its approach to generate expression parameters. SadTalker institutes a two-stage video generation model, designating 3DMM parameters as intermediate outputs. We capitalize on its first phase and employ it as our expression parameter generator G_e .

To elucidate further, guided by SadTalker’s computational pipeline³, the auditory input A undergoes a transformation to yield a mel-spectrogram sequence $M_{1:T}$, synchronizing with the total number

of video frames. Each frame represented as M_i exists within the dimensional space $\mathbb{R}^{80 \times 164}$. The sequence initiates with the expression parameter β_0 , while the subsequent computed facial expression parameters are articulated as $\hat{\beta} = G_e(\beta_0, M_{1:T})$.

3.4 Rendering Network R

The selection of the rendering mechanism significantly influences performance indicators as well as visual perception. Following the comparison of outputs derived from various renderers (comprehensive data results are accessible in our ablation study), we ultimately elect to implement the second-phase methodology from SadTalker for our rendering process. This procedure can be mathematically represented as $\hat{I}_i = R(I, \hat{\beta}_i, \hat{\rho}_i, \beta_0, \rho_0)$, where I designates the reference image and $\hat{I}_i$ signifies the i -th inferred image.

4 EXPERIMENTS

4.1 Datasets

The ViCo dataset [31], a compilation of videos, is predominantly sourced from the acclaimed video-sharing platform, YouTube. Characterized by its eclectic assortment of face-to-face dialogue clips, this dataset emphasizes the bidirectional information interchange between paired interlocutors.

Each of these video segments has been subjected to rigorous scrutiny and subsequent annotation by a specialized production team. The dataset allocated for the ViCo 2023 talking head generation track encompasses 440 speaker videos, each paired with its corresponding audio, exhibiting durations spanning from a brief 1 second to an extended 1 minute.

We have segmented this data into training and test subsets, adhering to a 9:1 distribution ratio. Every video clip has been processed to a resolution of 256x256 pixels and exhibits a rate of 30 frames per second. The finalized talking head generation track test subset, as disseminated by the ViCo Challenge, consists of 45 benchmark images coupled with their 45 pertinent speech audio recordings.

⁴Given the limited scale of the ViCo competition dataset, we did not retrain the SadTalker model and directly utilized the pre-trained model available in the open source. Consequently, we maintained the same input information as SadTalker. Hence, only the mel-spectrogram was extracted from the audio, instead of using the same input as the head pose generator.

³<https://github.com/OpenTalker/SadTalker>

4.2 Evaluation Metrics

To assess the effectiveness of our model, we adhered to the evaluation metric stipulated by the challenge. The quality of the resulting videos will be scrutinized quantitatively, taking into account visual quality, naturalness, and task-specific objectives.

Metrics under the purview of visual quality comprise CPBD [11], FID [7], SSIM [22], PSNR, and CSIM [3]. Specifically, CPBD calculates image sharpness; FID measures the disparity between synthetic and real data distributions; SSIM evaluates the perceptual similarity of image contrast, luminance, and structure; PSNR quantifies the synthetic data's error relative to real data; and CSIM gauges identity preservation using ArcFace.

Regarding naturalness, the incorporated metrics encompass ExpL1 and PoseL1. Notably, ExpL1 measures the L_1 distance of expression coefficients derived from the 3DMM reconstruction, while PoseL1 calculates the L_1 distance of translation/rotation coefficients originating from the 3DMM reconstruction. For speaker-specific factors, we employed metrics such as LipLMD, AVOffset, and AVConf as per SyncNet [2]. AVOffset determines the synchronization offset of lip motion relative to the input audio, AVConf evaluates the confidence of AVOffset, and LipLMD measures the lip landmark distance.

4.3 Implementation Details

In training the head pose parameter generator G_p , we faithfully adhered to the parameter configurations delineated in prior work [10]. Specifically, a learning rate of 2×10^{-3} was adopted and the batch size was designated as 256 by default. It is imperative to highlight that, due to the absence of accessible training code for SadTalker, we directly utilized its expression parameter generator G_e and the associated rendering network R strictly for inferential operations. Across all experimental endeavors, the hyper-parameters were consistently set as: $\lambda_1 = 1$, $\lambda_2 = 5$, $\lambda_3 = 10$.

4.4 Quantitative and Qualitative Performance

As evidenced by Table 1, our methodology takes the lead in half of the evaluated metrics (CPBD, FID, PoseL1, AVConf, LipLMD), also achieving the most substantial overall score. These outcomes indicate that our method excels not only in image quality but also in the portrayal of head poses and facial expressions, particularly in terms of audio-visual synchronization. Several factors contribute to this superior performance. Predominantly, the effective amalgamation of the natural head pose parameter generator and precise expression parameter generator enhances the precision of predicted head poses and facial expressions. Furthermore, the implementation of the second-phase method from SadTalker as our renderer improves the overall quality of the generated images, resulting in significant enhancements in audio-visual synchronization. Our scrupulous consideration of inter-frame smoothness also leads to more natural and fluid motion in the final video output.

As presented in Figure 2, we exhibited the results corresponding to the 30th, 60th, and 90th frames. The figures unequivocally illustrate that our generated results not only retain a substantial amount of image details but also exhibit quite natural head poses. These observations harmonize well with our quantitative metrics, reinforcing the effectiveness of our model in generating high-quality synchronized videos.

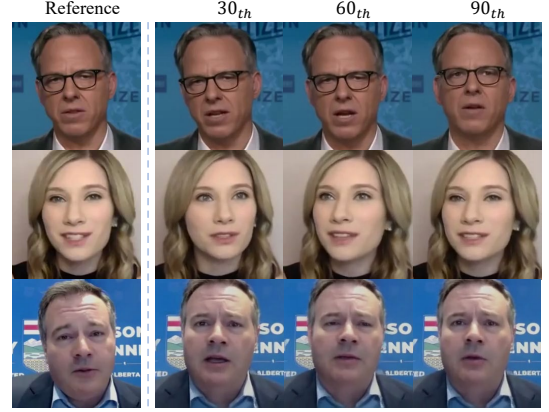


Figure 2: Qualitative evaluation of our model demonstrated on three distinct samples.

4.5 Ablation Study

As outlined in Table 2, we conducted a series of ablation studies to evaluate the importance of the chosen modules: 1) the effect of the head parameter generator, and 2) the significance of the renderer selection. However, due to the limited size of the ViCo dataset, it is not apt for retraining to predict facial expression parameters. As a result, we did not perform ablation experiments for facial expression parameters. We devised two variations of our model: 1) **w/o G_p** : we excluded our pose prediction network G_p and substituted it with the original pose prediction network from SadTalker. And 2) **w/o R** : we removed the renderer R from SadTalker and replaced it with PIRenderer[15]. Our module notably exhibits exemplary performance with respect to the head pose metrics, specifically *PoseL1*. Further, attributed to the mapping of sparse unsupervised keypoints, the renderer R consistently outperforms the configuration w/o R across an extensive array of visual quality metrics, including SSIM, CPBD, PSNR, and FID. For a comprehensive understanding of the network specifics related to the renderer R , we direct readers to SadTalker [26].

5 CONCLUSION

In this paper, we propose a two-stage framework for vivid talking head generation, making significant contributions to the field of one-shot talking head generation. Firstly, we introduce a natural head pose parameter generator and a precise expression parameter generator. These modules allow us to extract detailed and accurate 3DMM parameters, capturing the intricate dynamics of speaking behaviors. Secondly, we develop a robust rendering network that leverages the predicted 3DMM parameters to generate a series of vibrant images. Extensive experiments demonstrate the superior performance of our method in generating vivid talking head videos.

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